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PROMPT RADIATION AND MASS OUTFLOWS FROM THE STREAM–STREAM COLLISIONS OF TIDAL DISRUPTION EVENTS

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ABSTRACT

Stream–stream collisions play an important role in the circularization of highly eccentric streams that result from tidal disruption events (TDEs). We perform three-dimensional radiation hydrodynamic simulations to show that stream collisions can contribute significant optical and ultraviolet light to the flares produced by TDEs, and can explain the majority of the observed emission. Our simulations focus on the region near the radiation-pressure-dominated shock produced by a collision and track how the kinetic energy of the stream is dissipated by the associated shock. When the mass flow rate of the stream $\dot{M}$ is a significant fraction of the Eddington accretion rate, $\gtrsim$2% of the initial kinetic energy is converted to radiation as a result of the collision. In this regime, the collision redistributes the specific kinetic energy into the downstream gas and more than 16% of the mass can become unbound. The fraction of unbound gas decreases rapidly as $\dot{M}$ drops significantly below the Eddington limit, with no unbound gas being produced when $\dot{M}$ drops to 1% of Eddington; we find, however, that the radiative efficiency increases slightly to $\lesssim$8% in these cases of low $\dot{M}$. The effective radiation temperature and size of the photosphere are determined by the stream velocity and $\dot{M}$, and we find them to be a few times 10^4 K and 10^{14} cm in our calculations, comparable to the values inferred for some TDE candidates. The size of the photosphere is directly proportional to $\dot{M}$, which can explain its rapidly changing size as seen in TDE candidates such as PS1-10jh.

Key words: hydrodynamics – methods: numerical – quasars: supermassive black holes – radiative transfer

1. INTRODUCTION

A star can be tidally disrupted by a galaxy's central supermassive black hole (SMBH) when it passes within the tidal radius $r_t \equiv R_*(M_h/M_*)^{1/3}$, where R_* and M_* are the star's radius and mass, and M_h is the mass of the black hole. The candidates for tidal disruption events (TDEs) are usually found by searching for flares co-located with the galactic nucleus in the X-ray, UV, and optical bands (Komossa & Bade 1999; Donley et al. 2002; Gezari et al. 2006, 2009, 2012; Bloom et al. 2011; van Velzen et al. 2011; Arcavi et al. 2014; Holoien et al. 2016; Komossa 2015; Lin et al. 2015; Vinkó et al. 2015, and references therein). The flares typically reach a peak luminosity $\sim 10^{43}$–10^{44} erg s^{-1} within a few months, which corresponds to the Eddington luminosity of a $\sim 10^5$–$10^6\,M_\odot$ black hole.

The flares of TDEs have been thought to be produced from accretion disks that form after the stellar debris circularizes (Rees 1988; Strubbe & Quataert 2009). For the stellar stream with a flat energy distribution per unit mass, the mass-return rate decreases with time as $\propto t^{-5/3}$ (Phinney 1989). It is usually assumed that the stellar stream can be circularized and accreted quickly so that the observed luminosity also declines with time as $\propto t^{-5/3}$, and this has been adopted as a hallmark of many observed TDEs. However, the effective thermal temperature of the observed flares is only a few $\times 10^4$ K and remains approximately constant when the luminosity declines (Gezari et al. 2012; Arcavi et al. 2014). This is difficult to explain if the flares are indeed produced from the standard accretion disks, whose temperatures should vary with varying rates of accretion (Beloborodov 1999).

Significant progress has been made recently in studying the detailed processes in TDEs based on grid-based or smooth-particle hydrodynamic simulations (Rosswog et al. 2009; Guillochon & Ramirez-Ruiz 2013; Guillochon et al. 2014; Hayasaki et al. 2016; Sadowski et al. 2016; Shiokawa et al. 2015; Bonnerot et al. 2016). These studies raise more challenges in the classical picture of TDEs. The simulations usually find that it is very difficult to circularize the stellar streams if the orbits remain parabolic, as might be expected when the orbital dynamics are Newtonian. Additionally, the distribution of mass per unit energy is not exactly flat (Lodato et al. 2009; Guillochon & Ramirez-Ruiz 2013), and when observed at a single frequency (such as a photometric band), the light produced by an accretion disk resulting from a TDE may not follow the bolometric luminosity (Lodato & Rossi 2011). All these effects will make the radiation emitted by any formed accretion disks deviate significantly from the $t^{-5/3}$ law, in contrast to the observed light curves of TDE candidates, which appear to follow the fallback rate, even when observed in a single band.

Alternative mechanisms for describing the appearance of TDEs that can solve some of these puzzles include a hydrostatic envelope around the black hole emitting at the Eddington limit (Loeb & Ulmer 1997; Coughlin & Begelman 2014), possibly in conjunction with powerful disk winds that limit the accretion rate (Strubbe & Quataert 2009; Metzger & Stone 2016; Miller 2015). Lodato (2012), Piran et al. (2015), and Svirski et al. (2015) have proposed that a shock near the apocenter can also produce the observed TDE emission. However, there have been no quantitative calculations showing how the photons are produced by this mechanism and whether the emission is consistent with the observed properties of TDEs. As shocks caused by the stream–stream collisions clearly play an important role in converting the kinetic energy of the streams into thermal energy (Rees 1988; Kochanek 1994), we wish to study here in detail the structure of the shock when the stream first collides with itself and how this

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conversion proceeds. We also want to evaluate whether the stream–stream collision will naturally produce an emitting structure with a size of the order of hundreds of astronomical units and UV temperatures, as has been observed for the thermal TDE candidates.

For stellar streams on a parabolic orbit around a black hole, the kinetic energy is the same as the gravitational potential energy but of opposite sign. When a non-negligible fraction of the kinetic energy is converted into thermal energy, the downstream gas is likely to be dominated by radiation pressure (Kim et al. 1999). All previous studies are based on hydrodynamic simulations with an assumed equation of state, and the thermal properties of the stream and the effects of strong radiation pressure are all neglected. In fact, the shocks are likely unresolved in most previous simulations. As Hayasaki et al. (2016) noted, the thermal properties of streams affect the circularization process significantly, and in this paper we consider streams that are initially "cold," in contrast to the wide fans that are realized only at low mass ratios (e.g., Rosswog et al. 2009; Guillochon et al. 2014; Shiokawa et al. 2015) or in deeply penetrating encounters (e.g., Sadowski et al. 2016). By using an accurate radiative transfer algorithm (Jiang et al. 2014b), we can study here the thermal properties of the radiation-pressure-dominated shock self-consistently for the first time.

The paper is organized as follows. The typical properties of the colliding streams resulting from tidal disruptions and how they collide with one another are described in Section 2. We describe the equations we solve and the numerical setup we use in Section 3. Our quantitative results, including radiative efficiencies and outflow rates, are provided in Section 4. In Section 5 we discuss the implications of our calculations and some directions for future study.

2. WHERE AND HOW STREAMS COLLIDE

When a star is disrupted by an SMBH, the debris stretches away from the black hole in a highly elongated stream. Despite recent assertions to the contrary (Krolik et al. 2016), cases where the star is not shock-heated during its closest approach at a pericenter distance $r_{\rm p}$ (with $\beta \equiv r_{\rm t}/r_{\rm p} \lesssim 3$, which constitutes 80% of tidal disruptions, Carter & Luminet 1985; Guillochon et al. 2009) result in post-disruption debris that is self-gravitating, with a width h that grows only modestly with distance from the black hole, $h \propto r^{1/4}$ (Kochanek 1994; Guillochon et al. 2014; Coughlin & Nixon 2015; Coughlin et al. 2016). In the simulation of Shiokawa et al. (2015), the self-gravity of the matter is explicitly switched off shortly after disruption, resulting in a debris stream that quickly broadens due to the stream's internal pressure gradient. This stream is artificially wider (by orders of magnitude) at the moment of self-interaction than in the more realistic case where self-gravity is included, with the end result resembling much rarer "deep" encounters in which the stream is broadened by the injection of heat at periapse. Even in cases where the stream passes through periapse multiple times before colliding due to out-of-plane precession effects, the streams remain much thinner than in simulations where self-gravity is ignored (Hayasaki et al. 2016). Because collisions between two thin streams are more likely to occur given the smaller rate of deep events, we ignore the less common fanned-stream scenarios. However, given that these deep events are likely to be associated with pre-disruption X-ray flashes (Guillochon et al. 2009) and environments with strong gravity that can produce detectable gravitational waves (Kobayashi et al. 2004), they deserve a full radiation-transport treatment as well, which we leave to future work.

For non-spinning black holes, the location of the first stream–stream intersection point can be determined (Dai et al. 2015) from the eccentricity of the most bound debris

$$e_{\rm mb} = 1 - 0.02(M_*/M_\odot)^{1/3}(M_{\rm h}/10^6 M_\odot)^{-1/3}\beta^{-1} \quad (1)$$

and the amount of instantaneous de Sitter precession at periapse (Guillochon & Ramirez-Ruiz 2015)

$$\Omega = \frac{6\pi G M_{\rm h}\beta}{c^2 q^{1/3} R_*(1+e_{\rm mb})}, \quad (2)$$

where $q \equiv M_{\rm h}/M_*$, with the distance of collision

$$r_{\rm c} = \frac{(1+e_{\rm mb})r_{\rm t}}{\beta[1 - e_{\rm mb}\cos(\Omega/2)]}. \quad (3)$$

Assuming $e_{\rm mb} \simeq 1$ and $\Omega \ll \pi$, the expression for $r_{\rm c}$ simplifies to

$$r_{\rm c} = \frac{16}{\Omega^2}\frac{r_{\rm t}}{\beta}. \quad (4)$$

Substituting the definition for Ω from Equation (2) and scaling to fiducial values, we find

$$\frac{r_{\rm c}}{r_{\rm s}} \simeq 12\beta^{-3}\left(\frac{R_*}{0.5R_\odot}\right)^3\left(\frac{M_{\rm h}}{10^7 M_\odot}\right)^{-2}, \quad (5)$$

where we have taken the fiducial stellar radius R_* to be equal to that of a 0.3 $M_\odot$ star, approximately the median mass of the standard Kroupa initial mass function (Kroupa 2001). Here $r_{\rm s}$ is the Schwarzschild radius, which is 2.63×10^{12} cm for a $10^7\,M_\odot$ black hole. As the streams move on nearly parabolic orbits, the speed of the fluid within the stream at the collision point is

$$\frac{v_{\rm c}}{c} = 0.3\beta^{3/2}\left(\frac{R_*}{0.5R_\odot}\right)^{-3/2}\left(\frac{M_{\rm h}}{10^7 M_\odot}\right). \quad (6)$$

The angle of collision is given by (Dai et al. 2015)

$$\theta_{\rm c} = \frac{1 - 2\cos(\Omega/2)e_{\rm mb} + \cos(\Omega)e_{\rm mb}^2}{1 - 2\cos(\Omega/2)e_{\rm mb} + e_{\rm mb}^2}, \quad (7)$$

which simplifies, again assuming $e_{\rm mb} \simeq 1$ and $\Omega \ll \pi$, to

$$\theta_{\rm c} \simeq \pi - \beta\left(\frac{M_{\rm h}}{10^7 M_\odot}\right)^{2/3}\left(\frac{R_*}{0.5R_\odot}\right), \quad (8)$$

which is $\simeq \pi - 1 = 123°$ for the fiducial values. $\theta_{\rm c}$ is typically larger than 90° except for the most relativistic encounters, meaning that the two streams interact violently at their point of intersection, converting a large fraction of the kinetic energies of both streams into heat.

2.1. Vertically Offset Stream Collisions

For spinning black holes, there is no a priori reason to expect the incoming star's orbital angular momentum vector to align with the black hole's spin vector, and thus the streams are likely to experience some degree of out-of-plane precession resulting from frame-dragging effects (Stone & Loeb 2012).

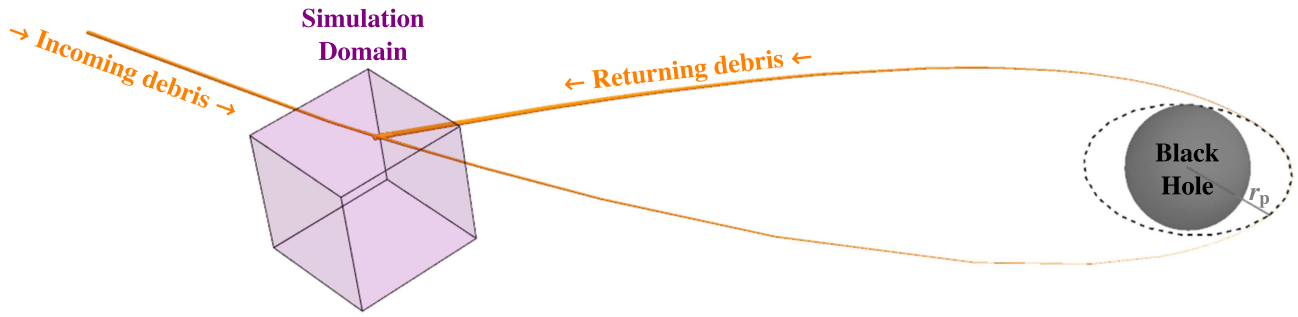


Figure 1. An example setup of our simulation domain with respect to the global geometry of the streams. The parameter combination shown here ($M_h = 10^7\,M_\odot$, $M_* = 0.18\,M_\odot$, $\beta = 0.64$, $a = 0.035$) is drawn from the Monte Carlo results of Guillochon & Ramirez-Ruiz (2015), and results in an angle of stream self-intersection of 138°.

The magnitude of this out-of-plane precession is

$$\Phi = \frac{4\pi a \sin i}{c^3 q^{1/2}}\left[\frac{GM_h\beta}{R_*(1+e_{mb})}\right]^{3/2} \quad (9)$$

where c is the speed of light, a is the black hole's dimensionless spin parameter, and i is the inclination of the star's orbit to the black hole's spin plane. Once the debris stream crosses periapse (the second time the matter crosses periapse after the initial disruption), the effects of self-gravity within the stream are minuscule and the stream's diameter is instead dictated by its internal pressure and the black hole's tidal gravity. If the stream's internal temperature remains low, as is expected for the highly eccentric disruptions that likely dominate the tidal disruption rate (Hayasaki et al. 2016), the stream's diameter when it returns to periapse can be even thinner than the star's original diameter. Multiplying this by r_c in Equation (5), we find a typical collision offset of

$$\frac{z_c}{D(r_c)} = 30a\beta^{3/2}\left(\frac{M_h}{10^7 M_\odot}\right)^{4/3}\left(\frac{R_*}{0.5R_\odot}\right)^{-3/2}\sin i \quad (10)$$

where $D(r_c) = 2R_*(r_c/r_p)$ is the stream's full diameter at the distance of the intersection r_c after exiting periapse (the stream marked "returning" in Figure 1), assuming that the stream expands homologously. For deep encounters of small stars about large, maximally spinning black holes, it is clear that the streams will frequently miss each other because the deflection angle is significantly larger than the size of the streams, in which case the onset of circularization may be significantly delayed, resulting in a "dark year" (Guillochon & Ramirez-Ruiz 2015). When the streams eventually do collide after the delay, they are more likely to do so close to the black hole at $r \sim r_p$, where they experience most of their precession. For grazing encounters of larger stars about lower-mass, lower-spin black holes,

$$\frac{z_c}{D(r_c)} = 0.05\left(\frac{a}{0.1}\right)\left(\frac{\beta}{0.5}\right)^{3/2} \times \left(\frac{M_h}{10^6 M_\odot}\right)^{4/3}\left(\frac{R_*}{R_\odot}\right)^{-3/2}\sin i, \quad (11)$$

and thus the offset can be much less than D, resulting in a nearly direct stream intersection.

The variety of possible angles, offsets, and collision locations as the black hole mass, stellar mass, and stellar size change suggests that the circularization process likely differs significantly from event to event. This motivates us to run separate simulations where these parameters are varied over a reasonable range. In this paper, we hold the location of the collision point fixed, and vary other parameters of the stream self-intersection, namely the angle of the collision, whether the stream collision is head-on or at some vertical offset, as well as a range of mass flow rates. Our numerical setup is described below.

3. NUMERICAL METHOD

3.1. Equations

We solve the frequency-independent (gray) radiation hydrodynamic equations in Cartesian coordinates (x, y, z) with unit vectors $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}})$. Our calculations employ the new radiation MHD code Athena++ (J.M. Stone et al. 2016, in preparation), which is an extension of the multidimensional MHD code Athena (Stone et al. 2008). The gravitational acceleration due to the central black hole is neglected, because we focus only on the collision point of the stream, with the simulation domain, particularly the region where most dissipation happens, being significantly smaller than the distance from the collision point to the black hole (see Figure 1). The timescale for the steady-state structure to be established after the collision is only ~1% of the orbital period. With the source terms for radiation momentum and energy, $\boldsymbol{S_r}(\boldsymbol{P})$ and $S_r(E)$, the equations we solve are (Jiang et al. 2014b)

$$\begin{aligned}
\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\boldsymbol{v}) &= 0,\\
\frac{\partial(\rho\boldsymbol{v})}{\partial t} + \nabla\cdot(\rho\boldsymbol{v}\boldsymbol{v} + P\boldsymbol{I}) &= -\boldsymbol{S_r}(\boldsymbol{P}),\\
\frac{\partial E}{\partial t} + \nabla\cdot[(E+P)\boldsymbol{v})] &= -cS_r(E),\\
\frac{\partial I}{\partial t} + c\boldsymbol{n}\cdot\nabla I &= S.
\end{aligned} \quad (12)$$

In the above equations, ρ, P, $\boldsymbol{v}$, and $\boldsymbol{I}$ are the gas density, pressure, flow velocity, and the unit tensor. The total gas energy density is $E = E_g + \rho v^2/2$, where $E_g = P/(\gamma - 1)$ is the internal gas energy density with a constant adiabatic index $\gamma = 5/3$. The gas pressure is $P = \rho k_B T/\mu$, where k_B is Boltzmann's constant, T is the gas temperature, $\mu = 0.62m_p$ is

the mean molecular weight for fully ionized gas of solar metallicity, and m_p is the proton mass.

The radiation source term S in the co-moving frame of the fluid is

$$S = c\rho\kappa_a\left(\frac{a_r T^4}{4\pi} - I_0\right) + c\rho\kappa_s(J_0 - I_0), \quad (13)$$

where κ_a and κ_s are the absorption and scattering opacities, while $J_0 = \int I_0 d\Omega$ is the angular quadrature of the specific intensity in the co-moving frame. The radiative transfer equation is solved with a similar method to that described in Jiang et al. (2014b), which has been successfully used to study super-Eddington black hole accretion disks (Jiang et al. 2014a), where the algorithm has been extended to include all v/c terms in ATHENA ++. The lab-frame specific intensity I is first transformed to the co-moving frame I_0 via Lorentz transformation and I_0 is updated implicitly with the source terms in the co-moving frame. The updated I_0 is then transformed back to the lab frame. All the velocity-dependent terms are handled automatically via Lorentz transformations, and the other parts of the algorithm are the same as in Jiang et al. (2014b). In the calculations presented in this work, we include only the electron scattering opacity, $\kappa_{es} = 0.34\,\mathrm{cm^2\,g^{-1}}$, and free–free absorption opacity, $\kappa_a = 2.86\times10^{-5}[\rho/(10^{-8}\,\mathrm{g\,cm^{-3}})]\,[T/(10^6\,\mathrm{K})]^{-3.5}\,\mathrm{cm^2\,g^{-1}}$, as in Jiang et al. (2014a). Effects of additional opacities caused by the metals, which are clearly important for the observational appearance of TDEs (Kochanek 2016; Roth et al. 2016) and the formed accretion disks (Jiang et al. 2016), will be studied in the future.

3.2. Simulation Setup

We inject two streams into the simulation box through the left x and y boundaries, which represent the self-collision part of the same stream (Figure 1), where the stream is assumed to be formed from a disruption of a low-mass star by an SMBH. Because the collision can occur at a variety of distances from the black hole at various velocities and angles, the setup we have chosen can simultaneously apply to very different physical systems, because many will coincidentally share the same conditions at the stream collision point. Figure 1 shows the size of the simulation domain relative to the disruption of an $M_* = 0.2\ M_\odot$ star that is tidally disrupted by an $M_h = 10^7\,M_\odot$ black hole with a periapse distance $r_p \simeq 2r_s$, for which the collision distance $r_c = 10r_s$, which gives the incoming stream velocity $v_i = \sqrt{2GM_h/r_c} = 0.31c$ for a stream initially on a parabolic orbit.

The typical scale height of the stream H is determined by the perpendicular component of the tidal force of the black hole at r_c and the gas pressure gradient (Guillochon et al. 2014)

$$H \approx 2r_c\sqrt{\frac{r_c}{r_s}\frac{k_B T_i}{\mu c^2}} = 7.37\times10^9\sqrt{\frac{T_i}{10^6\,\mathrm{K}}}\ \mathrm{cm}, \quad (14)$$

where T_i is the initial temperature of the stream before the collision. Notice that the scale height H is different from the stream's full diameter $D(r_c)$ defined in Equation (10). The full diameter $D(r_c)$ is the distance within which all the gas is contained, while H is the distance within which most of the stream's mass is contained. Only the scale height H is used to compute the velocity and density profiles. Hydrodynamic simulations of TDEs (Hayasaki et al. 2016) suggest that $T_i \sim 10^4$–10^6 K at the time of collision. Because the kinetic energy of the stream is 3×10^6 times the internal energy for $T_i = 10^5$ K, the exact initial temperature has negligible effects on the properties of the stream downstream of the shock after the collision. As there is no gravity in our simulations, H is a free parameter, and for numerical reasons we choose $T_i = 10^6$ K to set the thickness H so that the incoming stream is well resolved. The gas temperature of the incoming stream is chosen to be 10^5 K so that it is much colder than the downstream gas. The density profile of the stream is taken to be $\rho = \rho_0\exp(-0.5s^2/H^2)$, where s is the distance to the center of the stream and ρ_0 is the density at the center. We assume a velocity profile across the two streams that is mostly constant with $v = v_i$ but with a sharp decline at a distance $\gtrsim 2H$ from the stream center, $v = v_i\exp[-(s/2H)^{16}]$. This sharp cutoff is chosen to avoid numerical issues in the low-density region immediately exterior to the streams. The mass flow rate within each stream is $\dot{M}/\dot{M}_{Edd} \approx 2\pi H^2\rho_0 v_i/\dot{M}_{Edd} = 120\rho_0 r_s\kappa_{es}/\mathbb{C}^2$, where κ_{es} is the electron scattering opacity, $\dot{M}_{Edd} = 40\pi GM_h/(c\kappa_{es})$ is the Eddington accretion rate with an assumed $\epsilon = 10\%$ efficiency, and $\mathbb{C} \equiv c/\sqrt{k_B T_i/\mu} = 2.55\times10^3/\sqrt{T_i/10^6\,\mathrm{K}}$ is the ratio between the speed of light and the isothermal sound speed at a temperature T_i. The actual mass flow rate of streams from TDEs varies in a large range for totally and partially disrupted stars of different masses and stellar types (MacLeod et al. 2012; Guillochon & Ramirez-Ruiz 2013), and varies in time for a given event by many orders of magnitude. A typical value for the density at the stream center, given an Eddington ratio $\dot{M}/\dot{M}_{Edd} \sim 0.1$ for the example shown in Figure 1, is $\rho_0 \sim 10^{-8}$ g cm^{-3}, which is very optically thick ($r_s\rho_0\kappa_{es} \sim 10^4$) as a result of the mass being funneled through a narrow stream. In our simulations, we vary the mass flow rate to determine how our results change with changing fallback rates.

The streams will collide with angles usually in the range $\theta \sim 40°$–$160°$ due to relativistic precession (Kochanek 1994; Kim et al. 1999; Dai et al. 2015; Guillochon & Ramirez-Ruiz 2015). We choose two examples with $\theta = 135°$ and $90°$ to show the effects of varying the collision angle, which are labeled as A1 and A2. For a non-spinning black hole, the streams will collide in the orbital plane, and we have made this assumption in cases A1 and A2. With black hole spin, the precession is no longer confined to the orbital plane, which can cause a vertical offset when the streams collide (see Section 2). To treat this case, we apply a vertical offset $z_c = H$ to a simulation identical to A1 while keeping all the other parameters fixed, and label this as A1z. To study the cases with different injected mass flow rates, runs A1m1, A1m2, and A1m3 have mass flow rates that are 200%, 40%, and 10% of that in A1, while the other parameters are the same as in A1. The box sizes of all the simulations are fixed to be $L_x = L_y = L_z = 2r_s$ with resolution $N_x = N_y = N_z = 384$, and thus the two injected streams are initially resolved with 10 cells each across their diameters; once the streams collide with one another and expand downstream, the post-collision region is well resolved. The characteristic timescale is the stream crossing time $t_0 \equiv r_s/v_i = 3.19\times10^2$ s, which we will

Table 1
Model Parameters

Label	ρ_0	H	v_i	θ	z_c	β	$\log(M_h)$	a
Units	5.94×10^{-9} g cm^{-3}	7.37×10^9 cm	c		H		$M_\odot$	
A1	1	1	0.31	135°	0	0.64	6.95	0.035
A1z	1	1	0.31	135°	1	0.59	5.57	0.84
A2	1	1	0.31	90°	0	1.19	5.15	0.26
A1m1	2	1	0.31	135°	0	0.64	6.95	0.035
A1m2	0.4	1	0.31	135°	0	0.64	6.95	0.035
A1m3	0.1	1	0.31	135°	0	0.64	6.95	0.035

Note. Values of β, M_h, and a are example realizations from Guillochon & Ramirez-Ruiz (2015) that yield v_i, θ, and z_c similar to our simulation setup.

use as our fiducial time unit. The fiducial density unit is $\rho_0 = 5.94 \times 10^{-9}$ g cm^{-3} for all the simulations. We use 80 angles per cell for specific intensity I to resolve the angular distribution of the radiation field. The streams are injected at $x = -r_s$, $y = -0.72r_s$, $z = 0$ and $x = -0.72r_s$, $y = -r_s$, $z = 0$ for A1, $x = -r_s$, $y = -0.8r_s$, $z = 0$ and $x = -0.8r_s$, $y = -r_s$, $z = 0$ for A2. For A1z, the streams are injected at $x = -0.72r_s$, $y = -r_s$, $z = -0.5\,H$ and $x = -r_s$, $y = -0.72r_s$, $z = 0.5\,H$. Within $2H$ of the injection region, the ghost zones are fixed to be the initial condition of the incoming streams. In all the other regions of the six faces, we copy all the variables in the last active zones to the ghost zones, and we do not allow inflow except within the stream injection regions. Parameters of all the simulations are summarized in Table 1. Parameters of example realizations (β, black hole mass M_h, and spin a) of TDEs that can result in situations similar to our simulations are also listed in Table 1. All these simulations have the same scale height H and velocity v_i for the incoming streams. Simulation A1 differs from A1z only in the vertical offset z_c, and A1 differs from A2 only in the collision angle; these are designed to study the effects of spin and relativistic precession. Simulations A1, A1m1, A1m2, and A1m3 have the same parameters except for the central density of the incoming streams, and are designed to study the effects of different mass-return rates.

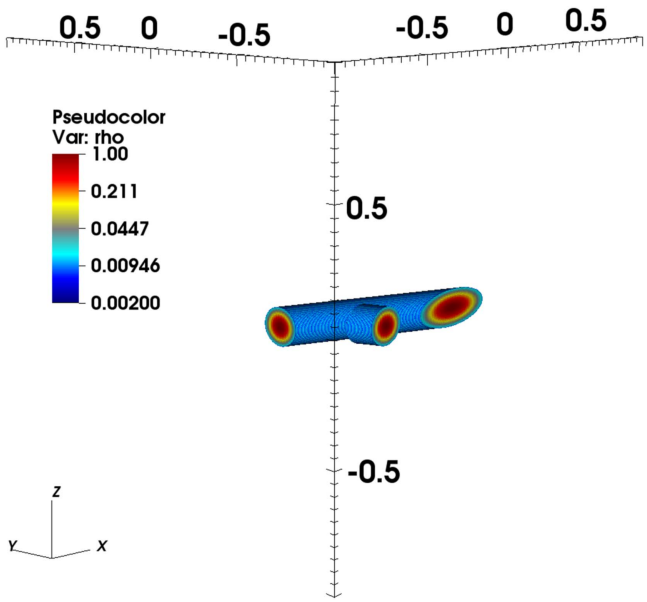


Figure 2. Density of the initial streams before the collision for run A1. The streams are injected from left x ($x = -1$) and y ($y = -1$) boundaries as described in Section 3.2, which represents the self-intersection part of the same stream from TDEs. The downstream gas leaves the simulation box through right x ($x = 1$) and y ($y = 1$), bottom and top boundaries. The unit of the box size is r_s and the density unit is $\rho_0 = 5.94 \times 10^{-9}$ g cm^{-3}.

4. RESULTS

4.1. Simulation History

The initial streams we inject from the boundaries as described in Section 3.2 for run A1 are shown in Figure 2. For all runs, the stream resulting from the disruption is injected from the left x boundary and is presumed to leave the domain from the right x boundary, returning to the domain through the left y boundary. The returning stream then intersects with itself at an angle θ, which is 135° for A1. The streams collide and a strong radiation-pressure-dominated shock is formed, which converts the stream's kinetic energy into thermal energy. The detailed structures formed in the post-shock region will be studied in the next section. Due to momentum conservation, the mean motion of the downstream gas is along the direction of the net velocity of the original two streams, which have the same density and magnitude of velocity. At the same time, the strong downstream radiation pressure accelerates the gas and causes it to expand. Most of the thermal energy is converted back to the kinetic energy within this expanding plume. The remaining thermal energy is emitted near the photosphere of the plume, where it can contribute to the light emitted by the TDE.

Histories of the density-weighted averaged kinetic energy density E_k, radiation energy density E_r, and gas internal energy E_g for A1 are shown in Figure 3. The thermal energies (E_r and E_g) initially increase rapidly while the kinetic energy drops due to the shock within a single stream crossing time t_0, with the radiation pressure being three orders of magnitude larger than the gas pressure. Afterwards, the system settles down to a steady state within a few t_0. The energy flux injected by the incoming stream is balanced by the outgoing energy fluxes carried by the downstream gas, with the density-weighted energy densities being dominant in the downstream region of the shock. The fact that E_r is comparable to E_k means that kinetic energy is efficiently converted to thermal energy in this case. The fluctuations in the steady state are caused by the unsteady shock front, which will be discussed in the next section.

4.2. Stream Structures after the Shock

A snapshot of 3D density ρ and radiation temperature T_r after the collision at time $29.6t_0$ of run A1 is shown in Figure 4. Here $T_r \equiv (E_r/a_r)^{1/4}$ is the effective blackbody temperature of the local radiation field. Due to symmetry, we split the domain in

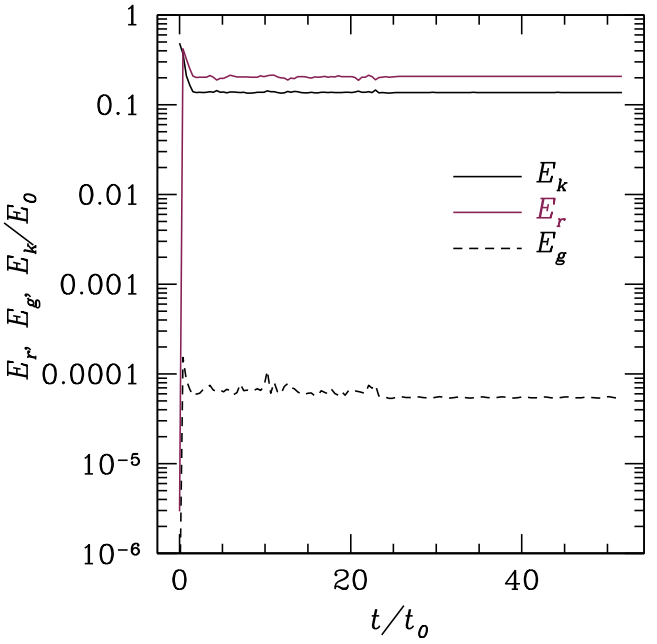


Figure 3. Volume-averaged, density-weighted kinetic energy density E_k (solid black line), radiation energy density E_r (red line), and gas internal energy E_g (dashed black line) as functions of time t for run A1. The unit used for the energy densities is $E_0 = \rho_0 v_i^2/2$ and the time unit is $t_0 = r_s/v_i = 3.2 \times 10^2$ s.

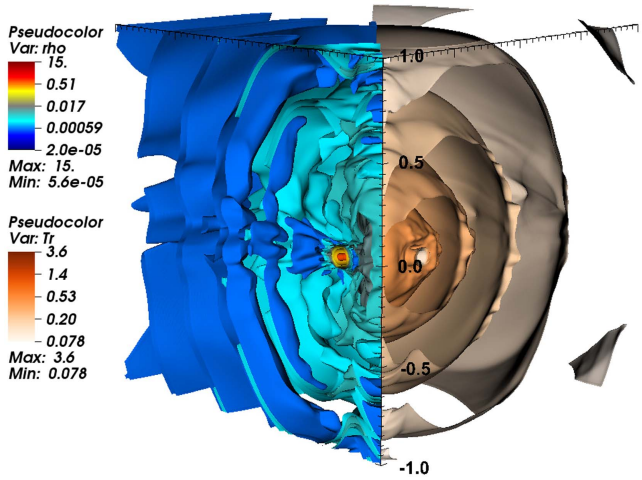


Figure 4. Snapshot of the 3D density ρ (left) and radiation temperature T_r (right) for run A1 at time $29.6t_0$. The density unit is $\rho_0 = 5.94 \times 10^{-9}$ g cm^{-3} and the temperature unit is $T_i = 10^6$ K. The length unit is r_s. (A movie showing the density evolution of this run is available at https://youtu.be/vc7O10UVFxg.)

two to show ρ and T_r simultaneously. The gas is redistributed in all directions after the shock as described in Loeb & Ulmer (1997). The nearly equal outgoing mass flow rates through all faces as listed in Table 2 confirm the roughly symmetric structure. To see the shock clearly, slices of density and flow velocity through the planes $z = 0$ and $x = y$ are shown in Figure 5. The high-density stream near $x = y = -0.8r_s$ is the location of the shock, and most of the gas flows within ~80° along the direction of the net velocity. Notice that a small fraction of gas is also injected in the opposite direction to the net velocity because the strong downstream radiation pressure accelerates gas in all directions. This is similar to the calculations done by Kim et al. (1999), although acceleration due to radiation was not included appropriately there, because they treated radiation as an effective isotropic pressure with a prescribed cooling function.

A high-density outflow is ejected perpendicular to the original orbital plane of the stream as shown on the left side of Figure 4, because the initial obscuration in the perpendicular direction is smaller and the acceleration due to the large downstream radiation pressure is larger. Despite the fact that the net mass flow rate has no perpendicular component, the flow direction with the largest velocity is ~10°–45° away from the original orbital plane $z = 0$. The shock front that develops between the two streams is unsteady and oscillates about the $x = y$ plane; as a result, high-density filaments form in the post-shock region, which are clearly visible in Figure 5. The periodicity of the instability appears to be of the order of the stream crossing time t_0, resulting in a distance between the filaments that is equal to the distance the flow travels within one oscillation period. The fluctuation of average energy densities shown in Figure 3 is also caused by the unsteady shock front. The physical reason that causes the shock front to oscillate was not initially clear, and to exclude numerical causes, we have checked that the instability exists as we varied the angle at which the streams were injected relative to the orientation of the grid (for example, one stream is aligned with the x axis while the other stream is at 135° with respect to the y axis). We then carried out a test hydrodynamic simulation of the collision where we set $\gamma = 4/3$, and the instability also existed with this softer equation of state (the shock is so optically thick that the photons provide an effective pressure with adiabatic index 4/3, so the presence of the instability for $\gamma = 4/3$ is not altogether surprising).

We speculate that the instability arises for the following reasons. When the two streams collide, the shock that develops between them has an angle that is initially equal and opposite relative to the two streams; this angle determines the obliqueness of the shock relative to the flows and thus the efficiency of conversion from kinetic to thermal energy. However, if a slight overpressure develops on one side of the shock, the shock "tips over" toward the opposite stream; this tends to make the shock less oblique relative to the other stream and thus results in more conversion of kinetic energy to internal energy on the opposite side of the shock. Now overpressured, the opposite side pushes back on the shock interface, tipping it back over in the opposite direction (making the angle with the first stream less oblique); the process then repeats itself ad infinitum as the shock angle tips back and forth on a timescale comparable to the sound-crossing time of the stream. When the optical depth drops with decreasing mass flow rate, the downstream photons can diffuse across the shock on a timescale shorter than the sound-crossing time, equalizing the pressure on the two sides of the shock. As a result, the overpressure from radiation is reduced, and the instability is significantly suppressed as seen in simulation A1m3 (see Section 4.5).

The downstream radiation temperature at the shock is 3.6×10^6 K and it drops to about 10^5 K near the boundary of the simulation box. An order-of-magnitude estimate of the temperature can be made based on energy conservation, where we equate the radiation energy density with the kinetic energy density, $a_r T_{sh}^4 \sim \rho_0 v_i^2/2$. Here T_{sh} is the downstream radiation temperature at the shock, and gas internal energy and upstream

Table 2
Mass Flow Rates through All Faces of the Simulation Box

Label	$\dot{M}_0/\dot{M}_{\rm Edd}$	$\dot{M}_{x1}$	$\dot{M}_{x2}$	$\dot{M}_{y1}$	$\dot{M}_{y2}$	$\dot{M}_{z1}$	$\dot{M}_{z2}$	$\dot{M}_{x1,u}$	$\dot{M}_{x2,u}$	$\dot{M}_{y1,u}$	$\dot{M}_{y2,u}$	$\dot{M}_{z1,u}$	$\dot{M}_{z2,u}$	$\dot{M}_u$
A1	0.11	16%	15%	16%	15%	19%	19%	3.3%	2.2%	3.5%	2.2%	1.7%	1.6%	14.5%
A1z	0.11	32%	11%	32%	11%	7.2%	7.2%	0.82%	0.0	1.3%	0.00	1.7%	1.7%	5.5%
A2	0.11	4.2%	27%	4.2%	27%	19%	19%	3.7%	3.0%	3.8%	1.2%	0.00	0.00	11.7%
A1m1	0.22	15%	15%	15%	15%	20%	20%	5.4%	1.7%	4.9%	2.6%	0.96%	0.89%	16.5%
A1m2	4.4×10^{-2}	15%	14%	14%	15%	21%	21%	2.8%	0.43%	2.8%	0.84%	0.00	0.00	6.9%
A1m3	1.1×10^{-2}	11%	15%	6.8%	15%	26%	26%	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Note. The second column is the injected mass flow rate carried with the stream scaled by the Eddington accretion rate as defined in Section 3.2. The next 12 columns are the outgoing mass flow rates through the left (with subscript 1) and right (with subscript 2) x, y, z surfaces, scaled by the total injected mass flow rate. The variables with subscript u represent mass flow rates carried with the unbound gas. The last column is the fraction of unbound mass through all six faces.

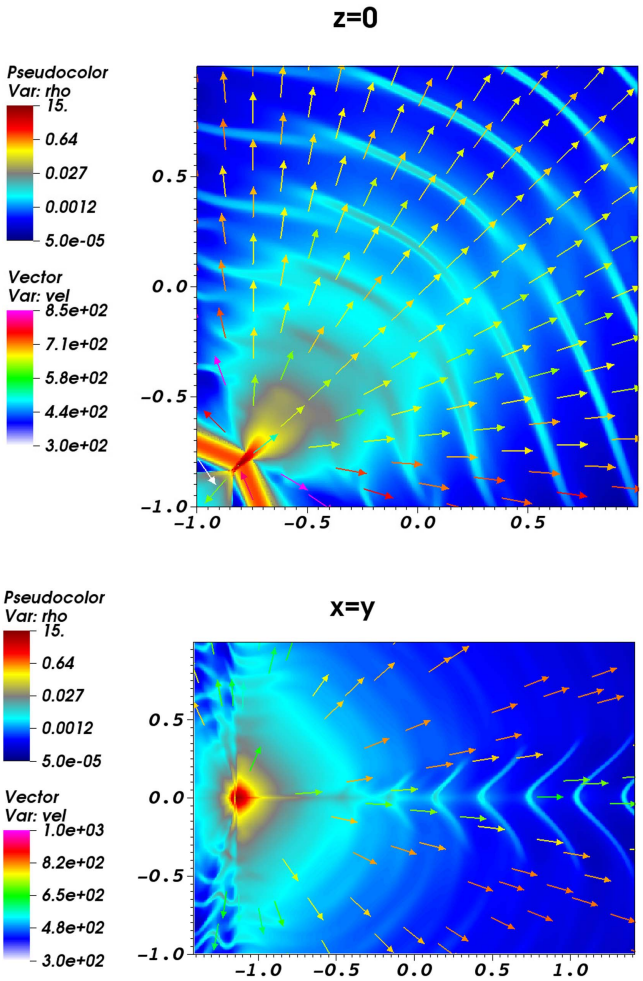


Figure 5. Slices of the density and velocity at $z = 0$ (top panel) and the plane $x = y$ (bottom panel) at time $29.6t_0$ for run A1. The density unit is ρ_0 and the velocity unit is 1.18×10^7 cm s^{-1}. The box sizes are in units of r_s.

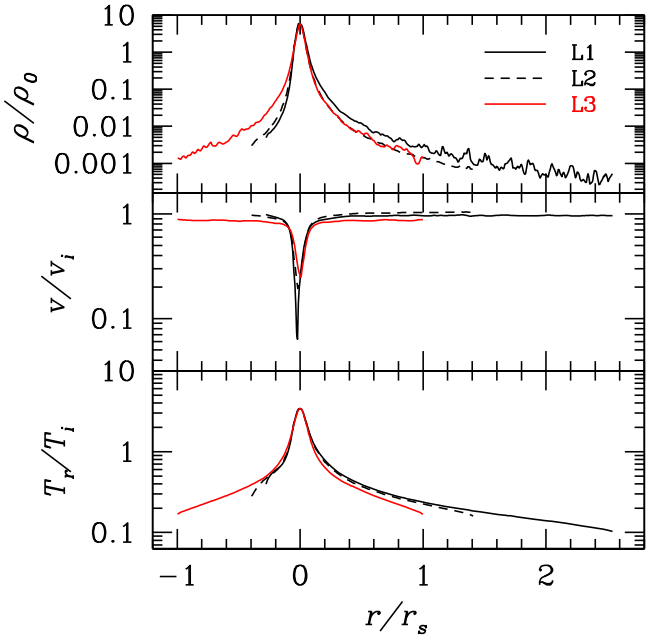


Figure 6. Time-averaged profiles of density (top panel), flow velocity (middle panel), and radiation temperature (bottom panel) along three different lines of sight for simulation A1. The time-averaging is done after the initial $10t_0$ (3.2×10^3 s). The three lines of sight are L1: $x = y$, $z = 0$, L2: $x + 0.8 = y + 0.8 = z/\sqrt{2}$, and L3: $x = y = -0.8$. The distance to the collision point $x = -0.8$, $y = -0.8$, $z = 0$ is r, where the sign represents the direction of flow velocity. All three lines go through the collision point $r = 0$.

thermal energies are neglected. For $\rho_0 = 5.94 \times 10^{-9}$ g cm^{-3} and $v_i = 0.31c$, $T_{\rm sh} \sim 4 \times 10^6$ K according to this approximation, which is similar to the temperature seen in the post-shock region. Comparing the gas pressure $P_{\rm gas}$ to the radiation pressure $P_{\rm rad}$, we also confirm that $P_{\rm rad} \gg P_{\rm gas}$ at the shock. Energy conservation implies that the downstream radiation temperature will increase with increasing mass flow rate for a constant v_i, but then the effective temperature at the photosphere will depend on how the photons are transported to the photosphere, which will be discussed in Section 4.5.

Time-averaged profiles of the density, flow velocity, and radiation temperature along three different lines of sight passing the collision point are shown in Figure 6. The line L1 is along the original orbital plane of the stream $x = y$, $z = 0$ while line L3 is the z axis at the collision point. The line L2 is at 45° with respect to the z axis in the $x = y$ plane. These lines of sight all probe the properties of the downstream gas. The flow velocity reaches the minimum value at the shock position, where the density and radiation pressure peak. The maximum downstream density is enhanced by a factor of 6 compared with the upstream density, which is consistent with the very strong radiation-pressure-dominated shock (Figure 13 of Jiang et al. 2012). The gas gets re-accelerated very quickly by the strong downstream radiation pressure and reaches a roughly constant velocity far away from the photosphere, where most of the thermal energy has already been converted back to kinetic energy. Density falls as r^{-2} when the velocity reaches a constant value. The velocities along L1 and L3 are always smaller than v_i, because most of the mass is injected along these

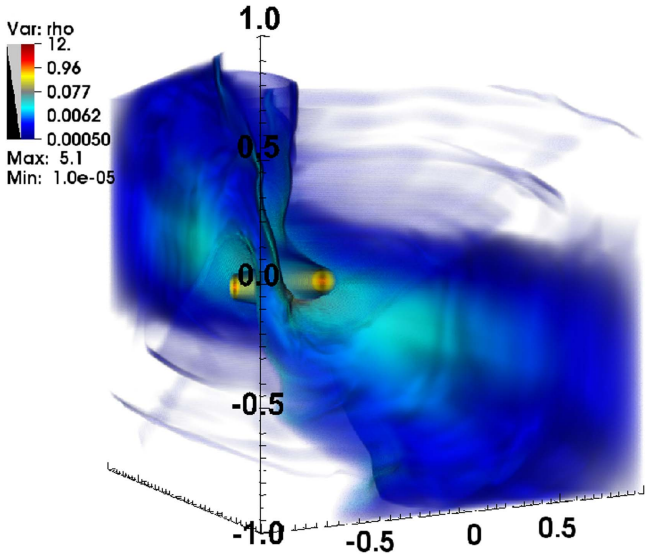


Figure 7. Snapshot of density at time $55t_0$ of the simulation A1z when a vertical offset is applied. The density unit is ρ_0 and the box size is in units of r_s. (A movie showing the density evolution for this run is available at https://youtu.be/jeLUixxE9rA.)

directions as shown in Figure 4. However, along L2, the flow velocity can be larger than v_i. This will be the source of most of the unbound gas created by the stream–stream collision, which will be discussed in Section 4.6.

4.3. The Effects of Vertical Offset

The spin of the black hole can cause a vertical offset of the streams when they collide (Section 2.1), which we study in simulation A1z. The setup is similar to A1 except that the centers of the two streams are located at $z = -0.5H$ and $z = 0.5H$. A snapshot of the density at $55t_0$ after the collision is shown in Figure 7. Because of the vertical offset and the exponential fall-off of density with distance from the center of the stream, the total momentum flux carried by the stream is much larger than the momentum flux of the gas involved in the collision. The primary result of the collision is instead a deflection of the two streams in the vertical direction, and less kinetic energy is converted to thermal energy in the shock than in A1. The radiation temperature of the downstream gas at the shock in A1z is 2.7×10^6 K, 32% of the radiation energy density in A1.

The vertical offset also causes anisotropic structures at the collision point. The gas that does not go through the shock is deflected from the original orbit, which remains cold. Table 2 confirms that mass flow rates going through the left x and y faces (where the streams would have exited the domain in the absence of a collision) are much larger in this case. Because of the large optical depth within the stream, photons cannot escape in this direction. Instead, most of the radiation is radiated along the direction perpendicular to the direction of travel of the deflected streams. The total optical depth from the photosphere to the collision point is also smaller, and is about 600 along the line of sight L2 compared with 2000 in A1 along the same line. As the downstream radiation energy density and the total optical depth are reduced by a similar factor (because they are both roughly proportional to the total amount of gas involved in the shock), the photospheric temperature along the direction in whiich photons can escape will be actually very similar to that in case A1. But because of the reduced photospheric radius and the solid angle where photons can escape—a region comparable to the sizes of the original streams—the radiation efficiency is significantly reduced.

4.4. Effects of the Collision Angle

The effects of the collision angles are studied in the run A2, where the streams collide at 90° and everything else held the same as in A1. Due to momentum conservation, the net momentum in this case is larger than in A1, which also means that the available kinetic energy that can be thermalized by the shock is smaller. We find that the downstream radiation energy density at the shock is 62% of the value in A1. Most of the downstream gas is confined to a narrow fan-like structure within 20° along the diagonal line of the $x-y$ plane, which is the direction of the net momentum. As shown in Tables 2 and 3, the radiation luminosity and the amount of unbound gas produced by the collisions are reduced compared with case A1. The case of a smaller impact angle is more likely to occur for deeper, more relativistic encounters (Equation (8)), so given our result of less conversion of the stream's kinetic energy into outflows and radiation, relativistic encounters are more likely to retain their mass if the streams collide with the same velocity v_i. However, if the velocity at the collision point is larger for relativistic encounters, for example when the collision point happens to be closer to the black hole, the outflow rate will be larger.

4.5. Effects of the Injected Mass Flow Rate

The four runs A1m1, A1, A1m2, and A1m3 cover mass flow rates from 22% to 1.1% of $\dot{M}_{\rm Edd}$; this progression could be seen to be representative of the declining phase of a TDE, which is the best-observed period given its prolonged duration (five years after disruption in the case of Gezari et al. 2015, for example). The geometry of the downstream gas is very similar for the four cases because they have the same collision parameters. But when the mass flow rate drops to 1.1% of $\dot{M}_{\rm Edd}$, the shock front becomes steady, with the instability we observed so prominently in A1 disappearing. A slice of density and flow velocity through the $z = 0$ plane for run A1m3 is shown in Figure 8. This structure establishes itself quickly after the initial collision, and the oscillation of the shock front as shown in Figure 4 does not appear here.

We take the line of sight L1 as an example to check the radial profiles of density ρ, radiation flux in the co-moving frame

$$F_{r,0} \equiv \sqrt{F_{r,0x}^2 + F_{r,0y}^2}, \tag{15}$$

and radiation temperature $T_{\rm r}$ as functions of the distance to the collision point; these are shown in Figure 9. The density is flat in the acceleration region and it declines with radius roughly as r^{-2} when the velocity almost reaches a constant value (see Figure 6 for A1). The profiles of density and radiation temperature are very similar in the four runs except for their normalization, which is proportional to the incoming mass flow rate. The co-moving radiation flux also declines with radius roughly as r^{-2}, except for run A1m3. The radiation energy density E_r declines with radius roughly as r^{-3}, which is consistent with the diffusion equation and the fact that $\rho \propto r^{-2}$ and $F_{r,0} \propto r^{-2}$.

Table 3
Kinetic and Radiative Energy Fluxes Produced by Collisions

Label	L_r	$L_{r,0}$	$L_{k,x2}$	$L_{k,y2}$	$L_{k,z1}$	$L_{k,z2}$	$L_{r,x2}$	$L_{r,y2}$	$L_{r,z1}$	$L_{r,z2}$
A1	3.5%	1.9%	22%	21%	27%	27%	0.65%	0.67%	1.1%	1.1%
A1z	2.4%	1.4%	10%	10%	6.9%	6.9%	0.36%	0.36%	0.84%	0.84%
A2	2.6%	1.4%	31%	30%	18%	18%	0.63%	0.61%	0.67%	0.67%
A1m1	3.2%	1.6%	22%	22%	27%	27%	0.61%	0.62%	0.99%	0.99%
A1m2	4.3%	2.4%	20%	22%	27%	27%	0.82%	0.84%	1.3%	1.3%
A1m3	7.7%	5.0%	19%	19%	27%	27%	1.7%	2.0%	2.0%	2.0%

Note. The first column is the total integrated radiation flux in the lab frame over right x, y, top, and bottom faces, while the second column is the total integrated radiation flux in the co-moving frame over the same four faces. The outgoing kinetic energy fluxes integrated over each face are listed in columns three to five while the other columns are the outgoing radiation fluxes in the lab frame integrated over each face. All these luminosities are scaled with the total radiation and kinetic luminosities in the lab frame from the four faces.

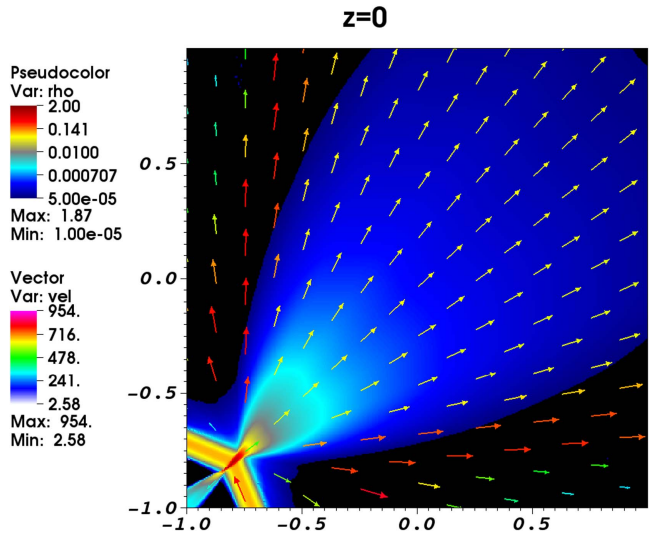


Figure 8. Slice of density and flow velocity in the plane of $z = 0$ at time $63.2t_0$ for run A1m3. The density and velocity units are ρ_0 and 1.18×10^7 cm s^{-1} while the box sizes are in units of r_s. (A movie showing the density evolution of this run is available at https://youtu.be/NYDUPy-0-2o.)

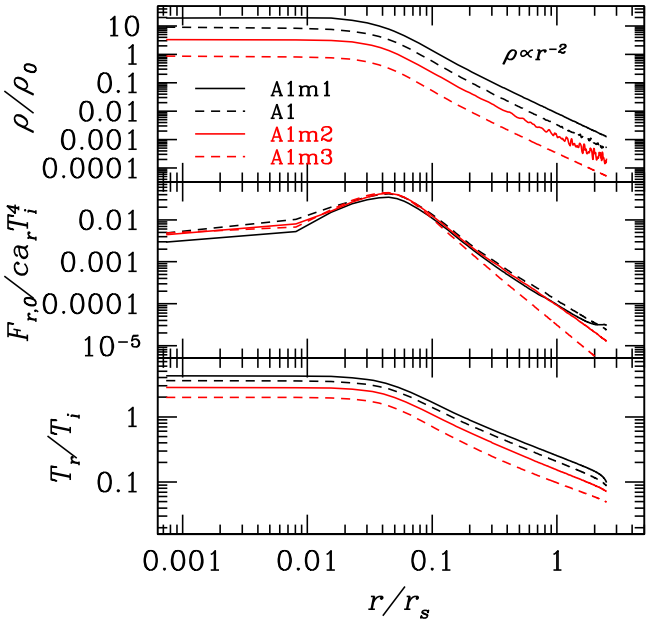


Figure 9. Top: radial profiles of the density along the line of sight L1 for runs A1m1, A1, A1m2, A1m3. The radius r is the distance to the collision point. Middle: radial profiles of the co-moving radiation flux along the same line of sight for the four runs. The radiation flux is scaled with $ca_rT_i^4$, where $T_i = 10^6$ K. Bottom: radial profiles of the radiation temperature T_r for the four runs.

The downstream radiation temperatures at the shock T_{sh} for the four runs are 4.2×10^6 K, 3.6×10^6 K, 2.8×10^6 K, and 2.0×10^6 K, which are consistent with our estimate based on energy conservation in Section 4.2. The radiation temperature in the optically thick part is determined by the diffusion equation,

$$\frac{\partial E_r}{\partial r} = -\frac{3\rho(\kappa_a + \kappa_s)F_{r,0}}{c}. \quad (16)$$

Here we have assumed the Eddington tensor to be $1/3\,\boldsymbol{I}$ for simplicity. If T_{sh} is much larger than the temperature T_{ph} at the photospheric radius r_{ph},

$$a_r T_{sh}^4 \approx 3a_r T_{ph}^4 \tau_{ph}, \quad (17)$$

where τ_{ph} is the optical depth from the collision point to the photosphere,

$$\tau_{ph} = \int_0^{r_{ph}} \rho(\kappa_a + \kappa_s)\frac{F_{r,0}}{a_r T_{ph}^4} dr. \quad (18)$$

Because the photosphere is not entirely within the domain in A1m1, A1, and A1m2, we will compare τ_{ph} integrated over a distance r_s instead of r_{ph} from the collision point (and denote it as τ_s) to compare between simulations. The flux-weighted optical depth τ_s and radiation temperature at r_s for A1m1, A1, A1m2, A1m3 are 2.8×10^4, 3.4×10^4, 4.3×10^4, 6.63×10^4 and 2.5×10^5 K, 2.0×10^5 K, 1.5×10^5 K, 9.6×10^4 K respectively. They all agree very well with Equation (17) and T_{sh} in the four runs. The ratios between the values of $a_r T_r^4$ at r_s for the four runs are very similar to the ratios of $a_r T_{sh}^4$, which are also the ratios of the incoming mass flow rate.

When the density follows $\rho_i r_i^2/r^2$, where r_i is a selected radius with density ρ_i, the total optical depth integrated from infinity to any radius r for electron scattering opacity is $\tau_{es} = \rho_i r_i^2 \kappa_{es}/r$. The photospheric radius can be estimated as[2] $r_{ph} \approx \rho_i r_i^2 \kappa_{es}$, which is $47r_s$, $24r_s$, $7r_s$, and $1.9r_s$ for the four runs along line of sight L1 respectively. Assuming that E_r continues to follow the scaling relation with radius as $E_r \propto r^{-3}$ and therefore $T_r \propto r^{-3/4}$, the photospheric temperature T_{ph}, which

[2] We take the radius where the electron scattering optical depth τ_{es} reaches 1 as the location of the photosphere.

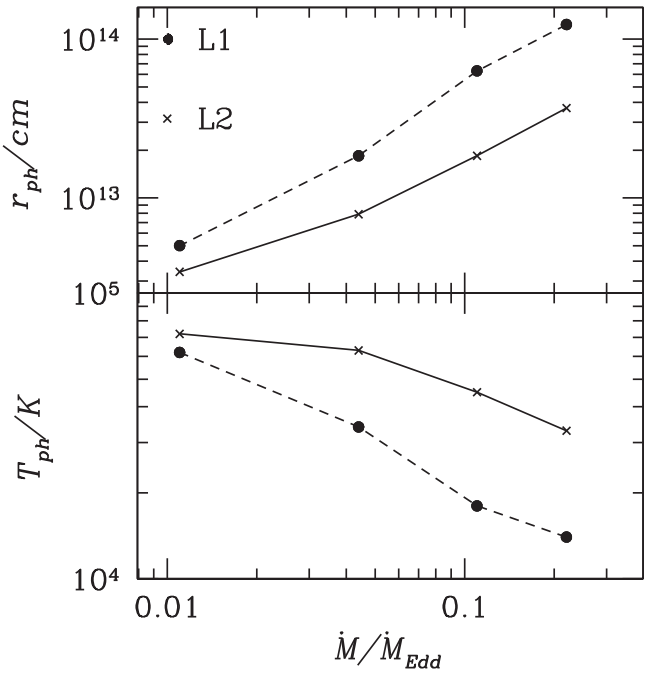


Figure 10. Electron scattering photospheric radius $r_{\rm ph}$ (top panel) and photospheric temperature (bottom panel) along lines of sight L1 and L2 as a function of injected mass flow rates for runs A1m1, A1, A1m2, A1m3.

is the radiation temperature T_r at $r_{\rm ph}$, along the line of sight L1 for A1m1, A1, and A1m2 is estimated to be 1.4×10^4 K, 1.8×10^4 K, and 3.5×10^4 K. For A1m3, the photosphere is captured in the simulation domain and the temperature is measured to be 6.2×10^4 K. A similar analysis along line of sight L2 shows that the photospheric radii are $14r_{\rm s}$, $7.0r_{\rm s}$, $2.9r_{\rm s}$, and $1.3r_{\rm s}$, while the photospheric temperatures are 3.3×10^4 K, 4.5×10^4 K, 6.3×10^4 K, and 7.2×10^4 K for the four runs. Notice that $r_{\rm s} = 2.6 \times 10^{12}$ cm for a $10^7\,M_\odot$ black hole.

The electron scattering photospheric radius and temperature as functions of mass flow rate for the four runs are summarized in Figure 10. For L1, a fit to the photospheric radius and temperature as functions of $\dot{M}$ shows that they are roughly proportional to the incoming mass flow rate as

$$r_{\rm ph,L1} \approx 6.5 \times 10^{14} \left(\frac{\dot{M}_0}{\dot{M}_{\rm Edd}}\right)^{1.09} \text{cm} \quad (19)$$

$$T_{\rm ph,L1} \approx 6.2 \times 10^{3} \left(\frac{\dot{M}_0}{\dot{M}_{\rm Edd}}\right)^{-0.51} \text{K}. \quad (20)$$

The above equations confirm that we only capture the photosphere for run A1m3. The scalings between $r_{\rm ph}$ and $T_{\rm ph}$ as a function of the mass flow rate along L2 when $\dot{M}_0 > 0.04\dot{M}_{\rm Edd}$ (we ignored the first point in our fit because the geometric effects change the relationship for lower $\dot{M}$) are

$$r_{\rm ph,\ L2} \approx 1.6 \times 10^{14} \left(\frac{\dot{M}_0}{\dot{M}_{\rm Edd}}\right)^{0.96} \text{cm} \quad (21)$$

$$T_{\rm ph,\ L2} \approx 1.8 \times 10^{4} \left(\frac{\dot{M}_0}{\dot{M}_{\rm Edd}}\right)^{-0.4} \text{K}. \quad (22)$$

Notice that $a_{\rm r} T_{\rm ph}^4 r_{\rm ph}^2$ from line of sight L2 is more than four times larger than the value along line of sight L1, which means more flux is emitted from the hotter surface nearer to the collision point, despite the reduction in surface area. Although the temperature will depend on the exact line of sight to the observer, the larger flux from the hotter line of sight suggests that a typical $T_{\rm ph}$ will tend closer to the values indicated by the expression for L2 (Equation (22)) than the expression for L1 (Equation (20)). For a given mass flow rate $\dot{M}$, as $a_r T_{\rm sh}^4 \propto \dot{M}^2/v_i$ and photospheric radius $\propto \dot{M}/v_i$, if we assume the same scaling relation for E_r, the photospheric temperature will change with stream velocity as $\sqrt{v_i}$. We note, however, that the observed photospheric temperature of TDE candidates should correspond to the radiation temperature at the photosphere of effective absorption opacity, which will be discussed in Section 5.

The total radiation and kinetic energy fluxes leaving from different faces of the simulation box are summarized in Table 3. The radiation luminosity from the top and bottom surfaces of the simulation box is larger than the radiation luminosity from the x and y faces except for run A1m3, because most of the mass is injected along the orbital plane of the original stream. This is consistent with the differences between lines of sight L1 and L2, which show that the photospheric temperature from the top and bottom surfaces is larger than the photospheric temperature from the x and y faces.

4.6. The Unbound Gas

The initial stream is in a parabolic orbit around the black hole, which means the kinetic energy is comparable to the gravitational potential energy. During the stream–stream collision, the kinetic energy is redistributed among the downstream gas via the shock. If one part of the gas receives more specific kinetic energy than the other parts, as shown in Figure 6, this part becomes unbound from the black hole while the others become more tightly bound. Because the black hole is not included in our calculations, we cannot evaluate the final fate of the bound and unbound gas (see Section 5). However, as kinetic energy is much larger than the thermal energy after the re-acceleration stops for the downstream gas, we can use the velocity alone to give a rough estimate of the fraction of unbound gas. If the flow velocity when the gas leaves the simulation box is larger than v_i, which by definition is at near-zero binding energy initially, we label it as unbound.

The mass flow rates carried with the unbound gas are summarized in Table 2. The fractions of the unbound gas for the four runs A1m1, A1, A1m2, and A1m3 are 16%, 14%, 6.9%, and 0% respectively. If all the unbound gas escapes to infinity, the kinetic energy luminosities carried by the unbound gas are 0.32%, 0.46%, and 9.8×10^{-5} of the total injected kinetic energy luminosity for runs A1m1, A1, and A1m2. This shows that the unbound gas will form only when the mass flow rate is larger than ~10% of $\dot{M}_{\rm Edd}$, and it increases with $\dot{M}$ above this threshold. If we consider only the gas leaving the top and bottom boundaries, which is unlikely to interact with the bound gas (which is mostly confined to the original orbital plane), the fractions are reduced to 2% and 4% for A1m1 and A1. As expected, we find that there is significantly more unbound gas produced with collision angle $\theta = 135°$ than there is with $\theta = 90°$. For the case with a vertical offset (A1z), less of the material is unbound than in the zero-offset case for the simple reason that a smaller fraction of the gas is involved in the collision, although a larger fraction of the gas is lost in directions perpendicular to the original orbital plane.

In all cases, the stream–stream collision clearly redistributes angular momentum within the two streams (Ramirez-Ruiz & Rosswog 2009). The downstream gas is redistributed in basically all directions, with only a small fraction of the gas remaining within the original orbital plane. However, without including the black hole in the simulation box, we cannot calculate the angular momentum exchange self-consistently. This will be the focus of future studies.

4.7. Radiation Efficiency

Although most of the radiation energy is quickly converted back to kinetic energy near the shock, a small fraction remains as radiation and the photons can escape the photosphere and contribute to the observed luminosity of a flare. The radiation produced in this way is directly connected to the initial kinetic energy of the stream without any time delay. Therefore, if the rate of mass fallback onto the black hole follows a given function, the stream–stream collision can produce a similar time dependence for the emitted radiation field, especially at early times when little matter has accumulated about the black hole from previously circularized material. At late times this could also reproduce the canonical $t^{-5/3}$ decay behavior expected for a large fraction of TDEs, but this may be subject to the additional complication that the black hole may have acquired a significant bound component (*à la* Loeb & Ulmer 1997) that may interfere with the stream collision process.

To see how efficiently the kinetic energy can be converted to the radiation energy during the stream–stream collision[3], we first calculate the total outgoing kinetic and radiation energy fluxes integrated over the right x, right y, top, and bottom faces; we do not include the left x and y faces because they are too close to the collision point. We define the efficiency as the ratio between the radiation flux and the total energy flux through the four faces, which are summarized in Table 3. For runs A1m1, A1, A1m2, and A1m3, the ratio between the total luminosity carried by the lab-frame radiation fluxes F_r and the total energy fluxes is 3.2%, 3.5%, 4.3%, and 7.7%. For runs A1m1, A1, A1m2, we do not capture the photosphere; this is just an upper limit for radiation efficiency. In principle a further fraction of the radiation could be converted back to kinetic energy before the photons escape to infinity. The radiation flux in the lab frame is the sum of the radiation flux in the co-moving frame $F_{r,0}$ and the advective radiation enthalpy $v(E_r + P_r)$ (Jiang et al. 2012), which can be converted to the kinetic energy flux within the photosphere. Here P_r is the component of radiation pressure along the direction of flow velocity v. Because $F_{r,0}$ is determined by the diffusion equation and it falls with radius as r^{-2} (Figure 9), the radiation luminosity carried by the diffusive flux will not be converted to kinetic energy, and this can be used to estimate the lower limit of radiation efficiency. We calculate the ratios between the total luminosity carried by the diffusive radiation fluxes and the total energy fluxes, which are 1.6%, 1.9%, 2.4%, and 5.0% for the four runs. This shows that the actual radiation efficiency varies between ∼2% and 7% and it increases weakly with decreasing mass flow rate.

[3] Notice that the radiation efficiency here is the fraction of the initial kinetic energy that is converted to radiation. This is different from the radiation efficiency in accretion disks, which is defined as the ratio of total radiation energy to the rest mass energy.

5. DISCUSSIONS AND CONCLUSIONS

The simulations we have performed are centered about the stream collision point, but as shown in Figure 1 the domain of the simulation may constitute a small fraction of the volume enclosed within the radius defined by the distance between the black hole and the collision. Gravity has been neglected in this work as we focused on the structure near the shock. As shown in Figure 6, the distance from the shock within which most of the radiation energy is converted back to kinetic energy is much smaller than r_c, and the inclusion of gravity is unlikely to play any role in changing this process. But gravity can influence the dynamics of the gas once it leaves the simulation domain; an example of a non-trivial interaction that can take place between the bound and unbound debris is shown in Figure 11, which was generated from our simulation A1m1, for which roughly 16% of the post-collision mass was estimated to be unbound. It is clear from inspecting the top view shown in Figure 11 that some of the bound debris, which will fall back toward the black hole eventually, lies on a track exterior to the path traveled by the unbound debris, and could impede the latter's escape, with the subsequent interaction resulting in a lower unbound fraction than what we have calculated. Even within the unbound debris, internal interactions between fluid elements of different velocities could result in additional energy dissipation within the outflow, as described in Strubbe & Quataert (2009) and Metzger & Stone (2016). Additionally, some of the material that is considered to be unbound has a trajectory that likely takes it close to the black hole's event horizon, possibly leading to direct accretion with little release of radiation (Svirski et al. 2015). These effects may lead to a lower unbound fraction than what we have calculated in Section 4.6.

As mentioned, our simulation domain is too small to capture the photosphere for our higher-$\dot{M}$ simulations. But as shown in Figure 9, our density profiles fall as r^{-2} from the collision point, suggesting that we resolve the outflow to the point at which the flow becomes homologous; this means that density isocontours should be self-similar beyond our domain (aside from the effects of the black hole's gravity). And because our highest-$\dot{M}$ simulation A1m1 shows a nearly isotropic ejection of matter, we do not expect the geometry of the outflow to change appreciably for accretion rates larger than what we have presented here. Under the assumption of homology, Figure 12 shows that at higher accretion rates the photosphere of the outflow produced from the stream collision point can completely envelop the black hole when $\dot{M} \sim \dot{M}_{\rm Edd}$. Once this global structure develops, the energy generated at the point of collision will be reprocessed along with radiation produced by the developing accretion disk. In these instances, the size of the photosphere may be sufficiently large to match the observed photospheric sizes of events like PS1-10jh, and additionally to match the observed rapid evolution of the size of the photosphere accompanied with increases and decreases in the accretion rate (see Figure 6 of Guillochon et al. 2014).

As an example, we consider the conditions that would be required in order to produce the $\gtrsim 2 \times 10^{44}$ erg s^{-1} observed at peak for PS1-10jh (corresponding to 15% of $L_{\rm Edd}$ for a $10^7\,M_\odot$ black hole) (Gezari et al. 2012). Given the 2% efficiency of converting kinetic energy to radiation that we find at the highest accretion rates, colliding streams with velocity $\sim 0.3c$ would need $\dot{M} \gtrsim 8\dot{M}_{\rm Edd}$ in order to produce the observed luminosity. We also note that we adopted the electron

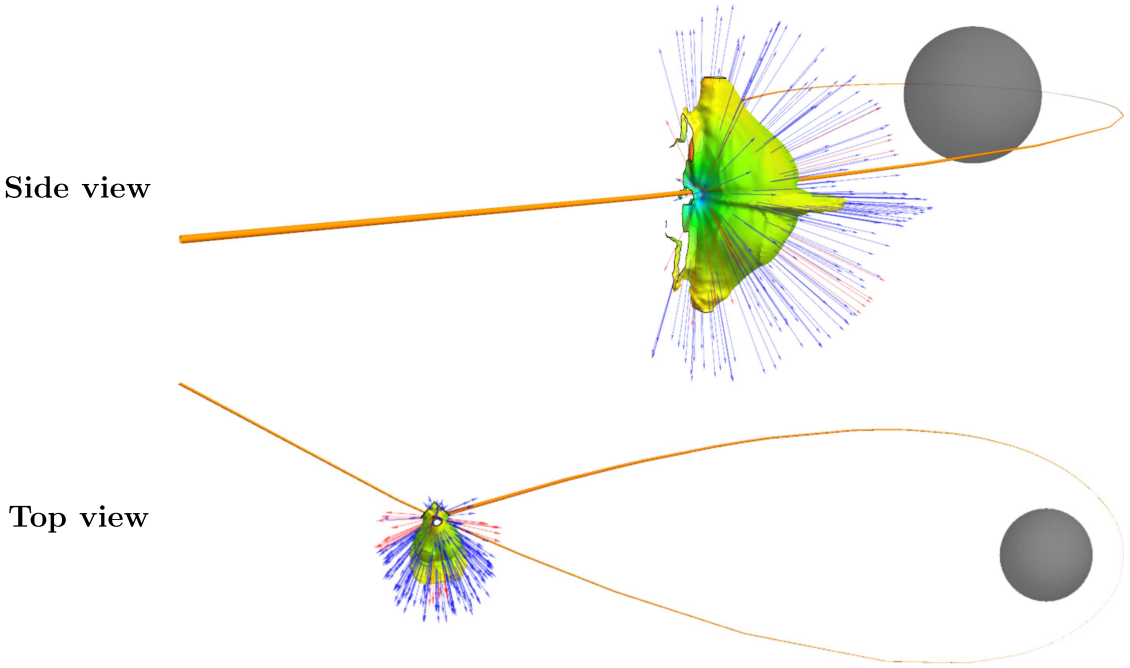


Figure 11. Three-dimensional rendering of simulation A1m1 where the simulation domain has been superimposed onto the global context that determined the stream collision conditions (as shown in Figure 1). The density isosurface shown corresponds to $\rho = 10^{-3}\ \rho_0$, with the color coding of the surface being proportional to the radiation temperature, ranging from blue (cold) to yellow (hot). The arrows emanating from the isosurface indicate the velocity vectors of the portions of the isosurface that they originate from, with blue arrows corresponding to material that is bound to the SMBH, and red corresponding to unbound material. It is clear that while the unbound debris may have the velocity required to escape the black hole, either its path may be impeded by bound debris or its trajectory may lead to direct accretion onto the black hole.

scattering and free–free opacity here for simplicity, but with temperatures above about several 10^4 K and density $\sim 10^{-8}$–10^{-9} g cm^{-3}, the additional opacity provided by the bound–bound transitions of iron-group elements can be much larger than the electron scattering value (Jiang et al. 2015). This additional opacity may be further supplemented by the bound–free transitions of hydrogen and helium (Roth et al. 2016). While the inclusion of these additional sources of opacity is unlikely to change the properties of the very optically thick region where the gas and photons were already tightly coupled, it will increase the total optical depth. As the observed photosphere should correspond to the location where the total optical depth for effective absorption opacity $\sqrt{\kappa_s \kappa_a}$ is one, when absorption opacity is larger than the electron scattering opacity, it will decrease the photospheric temperature relative to what is predicted by the simulations we present here. This could result in a larger photosphere that may potentially rival the distance to the black hole, which could potentially explain the large photospheric sizes observed for most optically selected TDEs.

Based on the 3D radiation hydrodynamic simulations we have presented in this work, we show that the shock completely changes the structures of the incoming stream and redistributes the kinetic energy among the downstream gas. When the mass flow rate of the stream is larger than 10% of $\dot{M}_{\rm Edd}$, this can cause a significant fraction of the incoming mass to have specific kinetic energy larger than the initial value and thus to become unbound from the black hole, while the remaining gas becomes more tightly bound. This process also converts 2%–8% of the initial kinetic energy into the radiation energy that eventually leaves the system as the observed flare. The kinetic energy luminosity carried by the unbound gas when it reaches infinity is $\lesssim$0.4% of the initial injection rate of kinetic energy, which corresponds to $\lesssim 5 \times 10^{42}$ erg s^{-1} when $\dot{M}_0 = \dot{M}_{\rm Edd}$ and $v_i = 0.3c$ for a $10^7\ M_\odot$ black hole. The photospheric temperature from the stream–stream collision is also consistent with the inferred photospheric temperature of optically discovered TDE candidates. As pointed out by Piran et al. (2015), the flares from stream–stream collisions can also follow the same time evolution as the mass flow rate, because there is almost no time delay between the collision and the production of photons. However, the stream collision would have difficulty in simultaneously producing X-rays and optical flux; while there is some variation in temperature along different lines of sight, the variation is not extreme enough to simultaneously produce optical and X-ray emission. This suggests that, for flares where both components have been observed, such as ASAS-SN 14li (Alexander et al. 2015; Holoien et al. 2015; Miller et al. 2015), a secondary process must be responsible for the X-ray emission if the stream collision is responsible for the optical/UV flux.

Because of the computational cost of executing hydrodynamical simulations with self-consistent radiation transfer, we chose to use a small box in order to fully resolve the post-stream collision shock region. As the size of the photosphere will be ~200 times larger than our current simulation domain when the mass flow rate reaches $\dot{M}_{\rm Edd}$, it is too expensive to simultaneously resolve the collision region and the photosphere within the same simulation at a fixed resolution. Additionally, we neglected to include the gravitational effects of the black hole, which will likely influence the overall shape of the photosphere, especially for higher accretion rates where the optically thick region encloses the SMBH itself. The gravity of the black hole will likely decrease the size of the photosphere and introduce interesting interactions between the envelope and possible accretion disks that may be formed afterwards. This will be the focus of future studies that account for global effects

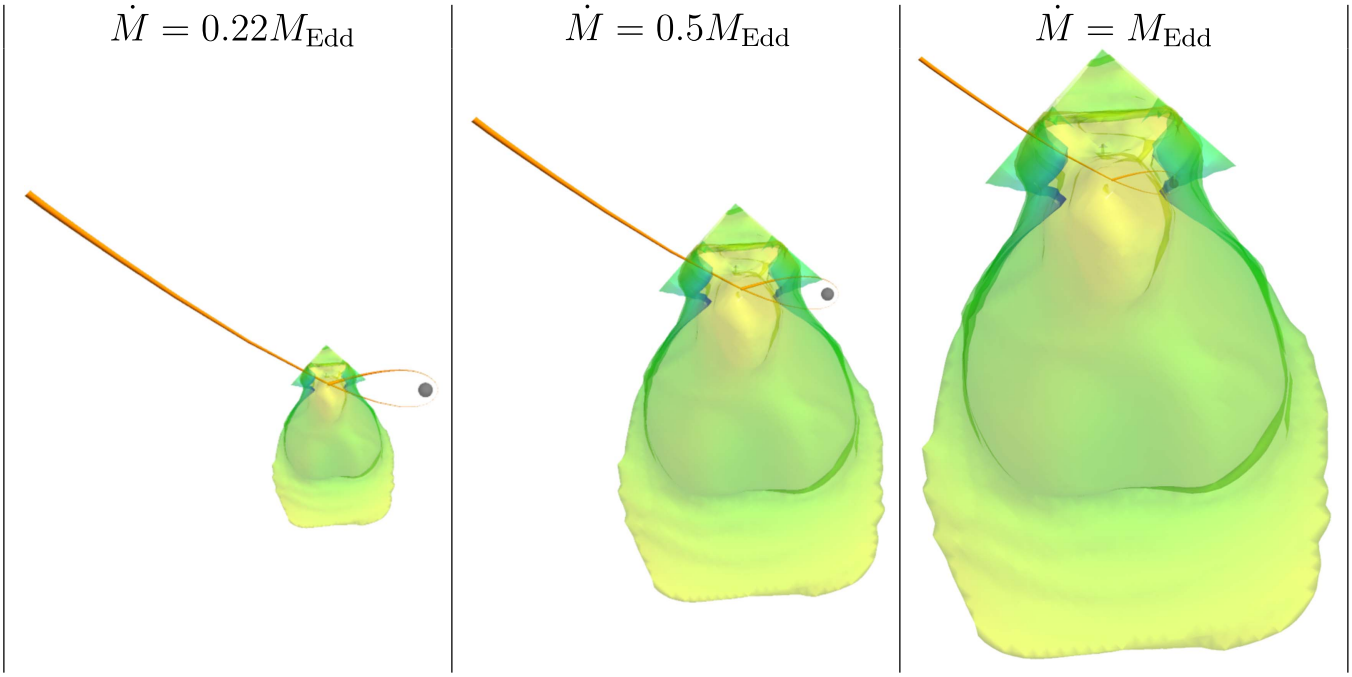


Figure 12. Photospheric geometry relative to colliding streams for $\dot{M} = 0.22 M_{\rm Edd}$ (corresponding to A1m1), $\dot{M} = 0.5 M_{\rm Edd}$, and $\dot{M} = M_{\rm Edd}$. In this figure we have taken the density contour evaluated at $\rho = 10^{-3}\,\rho_0$ (as shown in Figure 11) and scaled it to be equal in size to the photosphere, using Equations (19) and (21), resulting in $r_{L1,\rm ph} = 1.2 \times 10^{14}$, 3.1×10^{14}, and 6.5×10^{14} cm in the above panels (left to right). The stream diameter has been exaggerated by a factor of 5 in this figure to make it visible. At large accretion rates, the outflow from the stream collision point envelops the black hole, although we stress that the gravity of the black hole needs to be taken into account to determine the true global geometry of the photosphere.

on the resulting circularization structure. Lastly, after the original stream is destroyed by the collision, it will no longer return to the simulation box, at least not with a structure that is identical to the component of the stream that is returning to periapse for the first time. However, because the orbital period around r_c is $\sim 87 t_0$ where $t_0 \equiv r_s/v_i = 3.2 \times 10^2$ s, the steady-state structures within the stream collision region are established on a timescale that is much shorter than the time it takes for the stream collision angle to change due to deflection. This suggests that the collision process may be cyclical with a cycle period equal to roughly this period, resulting in times when the plume is being actively produced and other times when the two streams are not directly interacting (this effect can be seen to some degree in the highest-eccentricity simulations of Bonnerot et al. 2016). This effect is likely important to consider in the early phases of the stream–stream interaction, but evaluating the nature of this longer-timescale variability likely requires a global simulation.

In this paper we performed three-dimensional hydrodynamical simulations with self-consistent radiation transfer, which showed that a significant fraction of the kinetic energy carried by the debris streams can be released as observable radiation, with a photospheric size $r_{\rm ph}$ and temperature $T_{\rm ph}$ that depend directly on the rate of accretion $\dot{M}$, and typical values of $r_{\rm ph}$ and $T_{\rm ph}$ that are similar to observed optical/UV TDEs. As argued above, the full description of the stream collision process and its emergent radiation will likely require a global simulation to resolve, but the simulations we present here should capture most of the salient features associated with stream–stream collisions. While still under active development (see, e.g., Zhu et al. 2015), the addition of adaptive mesh refinement in ATHENA++ should allow these global-scale simulations to be performed in the near future.

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